Data Literacy and Research Data Management: The Croatian State of Affairs

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Abstract. This report is a part of an international survey on data literacy and research data management and is concerned with the Croatian state of affairs. A key aim of this study is to elicit the practices of Croatian researchers and PhD students regarding the production, dissemination, provision, storage, and description of research data, as well as portraying institutional attitudes towards those issues. By contributing Croatian data this study widens the international perspective on research data management. Findings of this study expose the current levels of awareness and gaps in knowledge and allow creation of structured and well-focused educational activities.

Keywords: Data literacy · Research data management · Academic staff
Croatia

1 Introduction

Openness is one of the key premises of contemporary research, which is reflected in many of its facets and processes, like access or diffusion, collaboration, evaluation etc. Although research and scientific knowledge are open by their very nature and intention, just the last decade brought this intrinsic notion to the fore, mainly owing to the availability of new digital technologies and collaborative tools. Conceptually, this premise has elicited a new frame of understandings labelled as "the fourth paradigm in scientific research" [1], which is characterized by new methods and instrumentation enabling scientists to gather huge volumes of data. This development leads to a new type of "data-intensive science" which shifts the attention from the end product of scientific work to documenting the decisions and context from which the research arises [2].

Implications of this data turn, or the "age of data," are described in many documents that reflect the current interest and hype among researchers and funding bodies, and in general at the societal and political level [3–5]. The narrative in these documents focuses on the benefits of open science, data sharing, and the necessity of creating an infrastructure entailing policies, services, tools, repositories, procedures that enable sharing, reusing, collaborating, curation, preservation etc.

However, for the concept of open science to succeed, it certainly takes more than the infrastructure. Willingness of researchers to share their data and use data of others is indispensable, and builds upon an open data culture which is not only anchored in

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attitudes, but also in competencies such as those related to data sharing, filtering, visualizing, linking, using, preserving, retrieving, that is, data literacy (DL). Thus, DL is, next to infrastructure, a key component of doing research in the age of data and data-intensive science.

Nonetheless, despite the impression that we deal with a new "version" of scientific work that brought about profound shifts and genuine new ways of understandings and doing things, as implied by the term "fourth paradigm", it actually represents a continuum in core values and principles in science (e.g., transparency, sharing of knowledge, repeatability, communication) that are accomplished by new approaches and different means. Continuity is also evident when it comes to DL, whose principles and content are standing in close connection to an existing narrative in research and education, namely information literacy (IL). Thus, DL can be perceived as an extension of IL applied within the research process [6]. According to some authors [7] DL has a chance to be more widely accepted by researchers than information literacy, because it is rooted in a research paradigm advanced by researchers, as evidenced by focus group interviews conducted by Colorado State University librarians among faculty members and researchers [8]. Additionally, as research policies and frameworks have nowadays put data-related procedures on their agenda (e.g., the creation of research data management [RDM] plans, the deposit of research data), understanding how to manage, navigate and curate huge amounts of data, that is, DL, has a very good chance to be accepted as a common requirement in research on institutional and system-wide levels.

Despite such potentials, the question of whether or not DL and data practices will be accepted by researchers depends on the existence of an open data culture, which is affected by different factors, as shown in [9]. This study implies that RDM practices and DL are very contextual phenomena, despite the fact that the notion of Open Science is inherently global. As the concrete context is obviously decisive for DL, insights into different contexts and determining contextual forces can shape and strengthen our understandings in DL. This paper is part of a multinational study and contributes to such insights into RDM practices and DL in Croatia.

2 Context of the Study

The effort to provide good support for RDM is a very current issue for Croatia. This is illustrated by projects and initiatives, like OPENAire[1] or SERSCIDA[2], which not only aim to establish a high-quality data infrastructure in Croatian higher education area, but implicitly emphasize the importance of research data management and RDM practices. For instance, OPENAire is focused on establishing the technical infrastructure for reliable digital asset management, while SERSCIDA addresses cooperation and knowledge exchange, but in the field of social science data archiving. Although in both projects infrastructural issues are in the spotlight, research data practices are also of interest as shown by SERSCIDA which elicited insights into research data practices

[1] https://www.openaire.eu/.

[2] http://www.serscida.eu/en/.

among social scientists based on a survey conducted in 2012 [10]. The study revealed that the data social scientists store is often very poorly documented and kept mostly on individual computers with several copies on different computers and media. Access to data is usually allowed only to project team members, but almost half of the respondents believed that their data should be made available either to the general scientific community or publicly available. Social scientists in Croatia consider it important to have access to research data collected by others, but in practice, secondary analysis is rare, and researchers often do not even know whether their data was used. The researchers also stated that data sharing was not part of the research culture.

In addition to projects, policies and statements of governing bodies are additional indicators of the heightened interest in research data. A good example that illustrates support for openness in research, RDM and, implicitly, DL is the recommendation issued by the Croatian Rectors Conference in 2015 [11], which promotes open access to scientific publication and research data and recommends that both open publications and research data should be considered in evaluation procedures and promotion of scientists, scientific projects, and other areas.

3 Objectives and Methodology

A key aim of this study was to elicit the practices of Croatian researchers and PhD students regarding the production, dissemination, provision, storage, and description of research data, as well as portraying institutional attitudes towards those issues.

The study was conducted at the University of Zagreb on a convenience sample of 515 members of academic staff and research students (7.3% of the population) distributed amongst six general subject areas.

The University of Zagreb is the largest university in Croatia employing 7100 researchers, which makes it one of the bigger European universities offering comprehensive access to higher education in almost every field of study. The survey constructed on the Limesurvey service was distributed via e-mail and consists of 24 questions. The data was collected in February, 2017. Data analysis focused on descriptive and frequency analyses.

The achieved sample, constructed based on voluntary participation, consists of two main groups of respondents: academic staff (82.3%) and research students (14.6%). From Fig. 1. it is evident that the respondents were mostly younger and middle-aged researchers, but an important thing to consider is that 83.7% have been involved in research for more than five years. With regard to their gender, 56.3% of respondents were female and 42.7% were male which resembles the distribution in the population. Our attempt to construct a representative sample by inclusion of a minimum of five percent of respondents from every scientific field showed that the interest for the researched topic may vary with regard to subject area.

Figure 2 shows the expected and achieved number (N) of respondents for each of the subject areas. We can see that the survey was very well received in natural sciences, the fields of engineering and technology, medical and health sciences, agricultural sciences and social sciences. However, we have not managed to spark the interest of researchers in the humanities to participate in this survey, even though they have twice

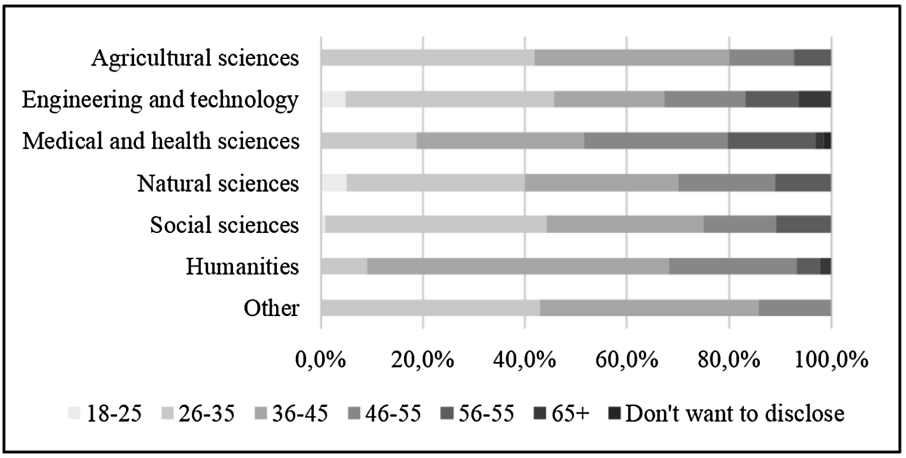


Fig. 1. Age of respondents by discipline affiliation

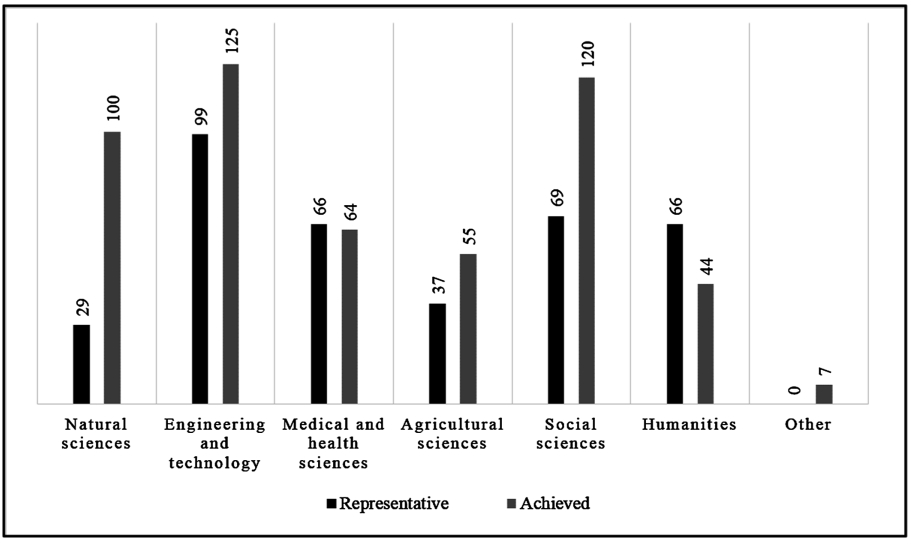


Fig. 2. Comparison of the expected minimum of respondents per field and the achieved number of respondents

as many employed researchers, as the natural sciences. Another important observation is the large number of respondents in the natural and social sciences, which greatly exceeds the expected minimum.

4 Results

The identity of the constructed samples requires that we consider disciplinary differences as an important factor of variability. Presented results will be considered in that sense wherever possible. Results can be categorized in the following way:

1. Type and volume of used and produced data
2. Usage, retrieval, storage and description practices
3. Attitudes towards data sharing
4. Perception of the role of institutions in RDM
5. Educational experience and willingness to participate in education about RDM.

4.1 Type and Volume of Used and Produced Data

In general, the data types most of our respondents use are standard office documents, images and web resources (Fig. 3).

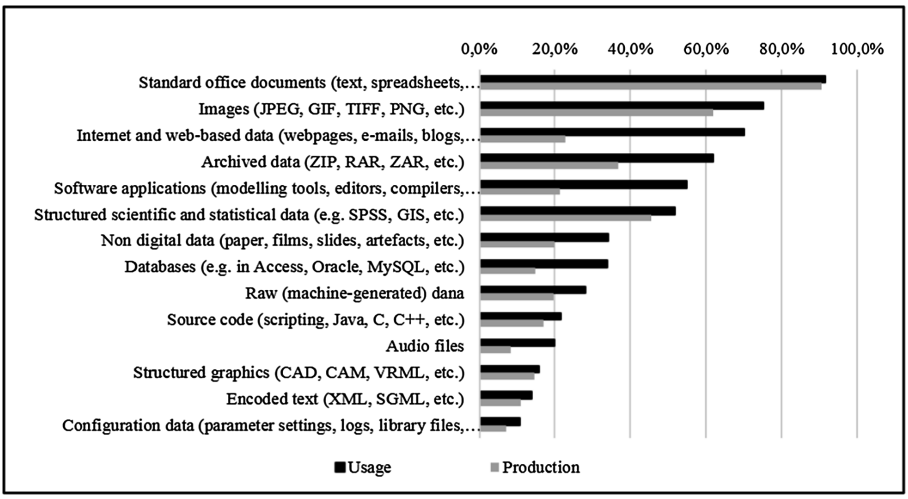


Fig. 3. Comparison of the usage and production of different data types

The production of any data type is, on average, lower then its usage. Differences between subject areas can be seen in higher usage and production of software applications and source code in natural sciences and the field of engineering and technology. Furthermore, differences are seen also in high usage of non-digital data in the humanities (70.5%) while usage of this data type does not exceed 35% in other fields relative to the number of respondents in a given field. Other interesting data types used and produced that were not mentioned by the survey were language corpora and video files.

The volume of the data respondents use (60.2%) and produce (54.37%) can mostly be measured in gigabytes (GB). Only 13.4% use, and 7.8% produce higher volumes of data which exceed one terabyte (TB).

4.2 Usage, Retrieval, Storage and Description Practices

The best insights into researchers' RDM practices come from understanding how they use data, where they retrieve the data they use in their research, and how they store and organize it. Respondents from the University of Zagreb mostly rely on own research skills so the main way they get data for their research are their own research projects (83.9%). On the other hand, 51.3% are prone to rely on their own research network and/or professional and personal connections to get the data they need.

53% use the data they get from outside sources as is, while 55% believe these resources still need a bit of cleaning and some modifications.

The data that 98.1% of respondents create or get from other sources is almost always stored on respondents' own devices. 45.8% also use cloud services, and 18.4% store it on central severs and institutional repositories, while 8.3% use outside repositories. Even though our respondents store the data at least on their own devices, when asked if they would like to store their research datasets beyond the lifetime of their projects, 48.5% said that they strongly disagree with the need for long-term storage, while only 30.1% agreed that it was important. Interestingly, even though only 18.4% store their data in institutional repositories 69.3% do believe that data would be best stored for long-term access at the university.

71.4% of respondents also believe that the university should pay for the storage and public access to the data our respondents created as its employees.

An awareness of the need for long term storage is also shown by description practices and assigning metadata to research datasets. Most of the respondents (55.7%) are familiar with the term "metadata" (uncertain = 15.5%; not familiar = 28.7%), and at least 60% assign some type of metadata to their research datasets. The metadata type added most often is administrative information (Fig. 4). However, when they were

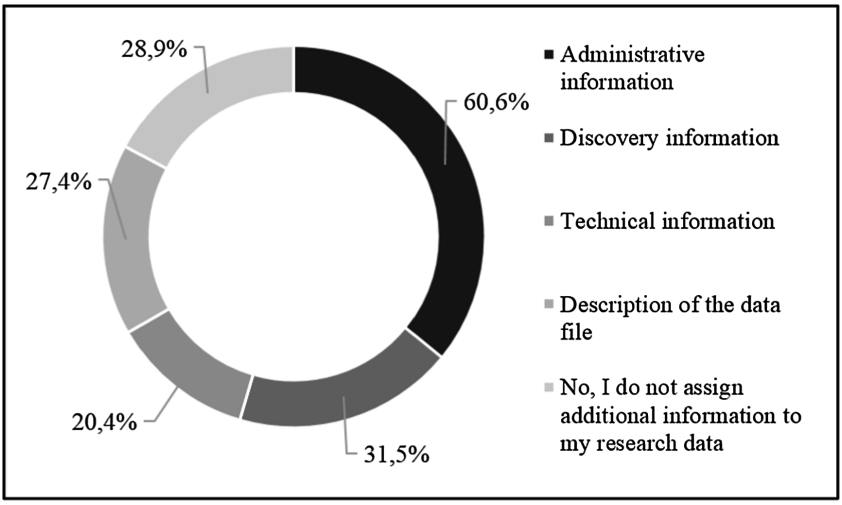


Fig. 4. Additional information respondents assign to their research data

asked how often they use a metadata standard for tagging their data 58.3% answer that they rarely do so. Usage, retrieval, storage and description practices of respondents did not vary across their different subject areas.

4.3 Attitudes Towards Data Sharing

The international survey pays special attention to data sharing practices. Figure 5 shows that respondents from the University of Zagreb usually do share their data and only a small percent say that they do not collaborate with other researchers. However, only 18.1% of that data is truly open to everyone.

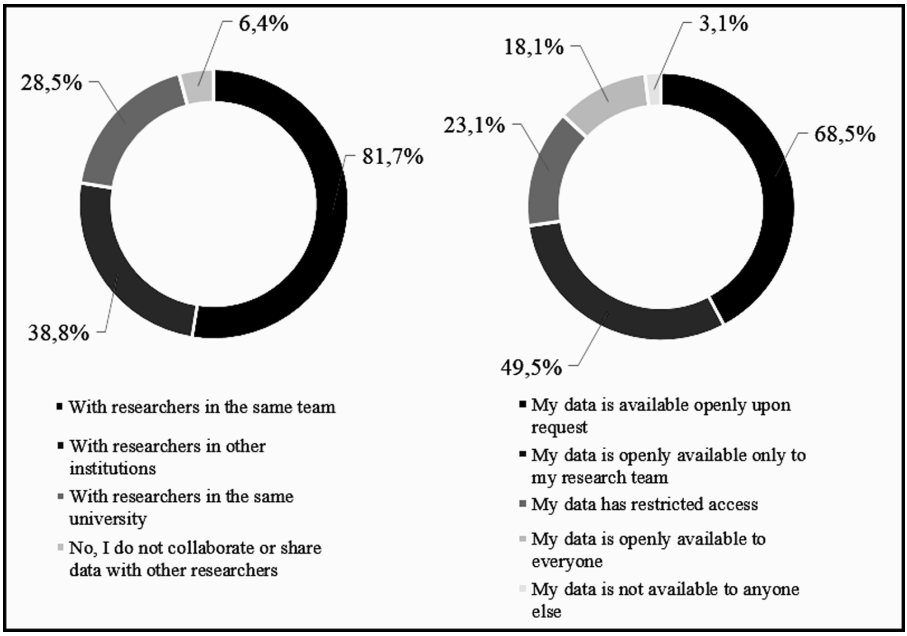


Fig. 5. Data sharing practices

We also asked our respondents about the concerns they have regarding the sharing of their data. With 50.5%, misinterpretation of data is the most common concern, and it is closely followed by misuse of data (48.7%).

However, all types of concerns were present to some extent (fear of losing the scientific edge, legal and ethical issues, lack of resources, lack of appropriate policies and rights protection). Discipline affiliation was not revealed as a contributing factor to the variability in data sharing practices.

4.4 Perception of the Role of Institutions in RDM

When they were asked to consider different institutional practices and how they perceive the role of institutions in RDM, respondents' answers mostly point to the perceived lack of institutional involvement or communication about RDM.

A specific focus in the survey was given to the concept of a Data management plan (DMP). From Fig. 6 we can see that respondents are mostly unsure whether their institution has such a plan, while 34.6% say that such a plan does not exist. Furthermore, other responses revealed that 81% say they have never used a DMP in their research projects and 78.4% say they do not have such a plan in their current research project. However, 35.5% say that they think such a plan would be useful while 59% remain uncertain about its usefulness. Understanding of institutional practices and roles as presented was comparable among different subject areas.

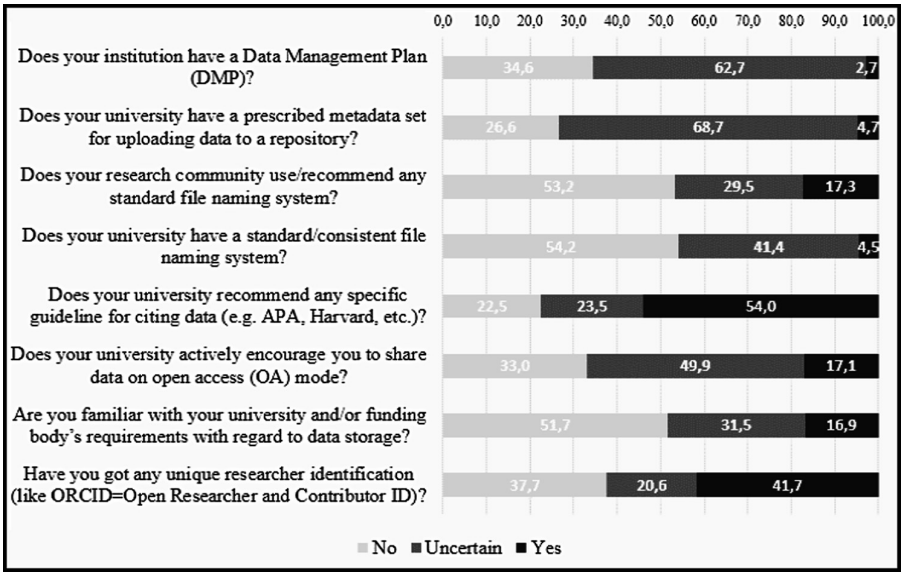


Fig. 6. Perception of the role of institutions and institutional practices

4.5 Educational Experience and Willingness to Participate in Education

One of the key aims of this survey was to establish the need for education and formal training about different RDM topics so that the research team could plan and provide activities to address these educational needs. While our respondents said that they did not have a lot of formal training on RDM, looking at the data (Fig. 7) we could say that their interest in these topics is moderate and similar among the different subject areas.

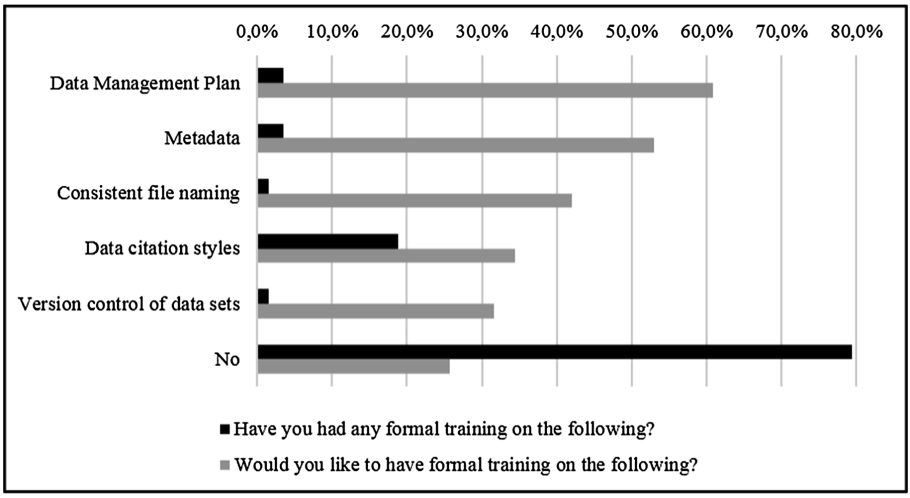


Fig. 7. Comparison of educational experience and readiness to participate in education on RDM

5 Concluding Discussion and Future Directions

This paper reports on a Croatian study which is a part of an international research project aimed at eliciting the RDM practices of European researchers in the higher education area. The study showed that researchers from certain subject areas exhibited high interest in participating in this survey, and we believe this should be considered as an incentive for further investigation. On the one hand, why do natural sciences and social sciences show an increased interest in the researched topic, while on the other, researchers from humanities show the opposite? One could discuss whether these differences come from different disciplinary foci and different disciplinary priorities that distinguish the subject areas. Do researchers in the humanities have as much to gain as do those from other areas by boarding the popular train going towards a data-intensive science? Disciplinary differences were also shown with regard to the data types used in a given field, but RDM and data sharing practices were common among all areas. This leads us to conclude that while the data content naturally differs, the common attitudes among the different subject areas lead to an understanding that there are areas of common interest which could be approached both on the level of institutional policies and educational initiatives focused on broadening the individual researchers' knowledge about RDM. The results given in this report corroborate the findings of previous research projects in Croatia because researchers still store their research data mostly on their own devices with an unsatisfactory level of quality regarding description and preservation. The data has shown that the academic community at the University of Zagreb shows awareness and understands the need to store, describe, share and manage research data but their data-related scientific practices do not reflect that understanding. The data sharing culture has yet to be fully developed because the willingness to share the data rarely takes the form of open public availability of research data. Furthermore, respondents' answers pointed out the perceived

lack of institutional involvement or communication about RDM which means that relevant institutions need to do more to encourage knowledge and skills acquisition about RDM. The reported low level of education on researched topics and an already existing moderate interest points to the need to develop DL programs that could shape the will of the academic community that promotes a systematic and sustainable approach to the management of research data. A lot of work for the data librarian!

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